Lecture 19

Noether's second theorem

A symmetry depending on an arbitrary function forces relations between field equations. Maxwell's identity in Lecture 9 is one example. We prove the general relation and identify its covariant form for a noncommuting gauge field.

Objectives. *We derive the formal-adjoint Noether identity for finite-order gauge operators. We apply it to pure Yang–Mills theory and compute Maxwell charge as an outward surface flux.*

By the end of this lecture you will be able to:

- Compute a gauge operator's formal adjoint, including coefficient derivatives.
- Derive the pure Yang–Mills covariant identity off shell.
- Distinguish ordinary and covariant Gauss laws and orient the charge flux.

19.1 Gauge parameters and compact variations

Let U be the interior of a compact region $D \subset \mathbb{R}^{1+d}$. Let $\phi = (\phi^1, \dots, \phi^N)$ be smooth on a neighbourhood of D. Assume its compact first variation has smooth finite-order Euler expressions $E_a[\phi]$:

$$\delta S_D[\phi]h = \int_D E_a[\phi]h^a \,\mathrm{vol} \qquad \big(h \in C_c^\infty(U;\mathbb{R}^N)\big).$$

These expressions may depend on derivatives of any finite order. For a multi-index $I = (I_0, \dots, I_d)$, write $|I| = \sum I_\mu$ and $\partial_I = \partial_0^{I_0} \cdots \partial_d^{I_d}$.

Definition 19.1 (Gauge operator and gauge symmetry). For a fixed integer $s_R \geq 0$, a *gauge operator* at ϕ is a linear map $R_\phi : C_c^\infty(U;\mathbb{R}^r) \to C_c^\infty(U;\mathbb{R}^N)$ given by

$$R_\phi(\varepsilon)^a = \sum_{|I| \leq s_R} R_\alpha^{aI}[\phi]\partial_I \varepsilon^\alpha. \tag{19.1}$$

Its smooth coefficients may depend on the point, the field and finitely many field derivatives. The variations are a *gauge symmetry* if $\delta S_D[\phi]R_\phi(\varepsilon) = 0$ for every compactly supported parameter ε, at every smooth field.

The coefficients are evaluated at the fixed field when the operator acts on the parameter. Their dependence on the field does not remove their ordinary coordinate derivatives in integration by parts. The compact parameter and its derivatives keep the variation away from the boundary of D.

19.2 The formal adjoint identity

Definition 19.2 (Formal adjoint and Noether identity)**.** The *formal adjoint* of (19.1), at the fixed field, is

$$(R_\phi^* E)_\alpha = \sum_{|I|\le s_R} (-1)^{|I|} \partial_I \big(R_\alpha^{aI}[\phi] E_a\big). \tag{19.2}$$

A differential relation among Euler expressions holding before imposing $E = 0$ is a *Noether identity*.

For example, in one coordinate, $R(\varepsilon) = a(x)\varepsilon_x$ has $R^*E = -\partial_x(aE)$, including the term $-a_x E$. Formal adjunction holds the field fixed but differentiates the evaluated coefficient as a function of the point.

Theorem 19.3 (Noether's second theorem)**.** *Under the preceding smoothness and first-variation hypotheses, a gauge symmetry gives the off-shell identity*

$$R_\phi^*(E[\phi])(x) = 0 \qquad (x \in U) \tag{19.3}$$

for every smooth field. This identity is asserted on U. No first-order assumption on the action or its Euler expressions is required.

Proof. Fix the field and move each parameter derivative by coordinate integration by parts. For each multi-index, repeated one-dimensional FTC gives

$$\int_D E_a R_\alpha^{aI} \partial_I \varepsilon^\alpha \,\mathrm{vol} = (-1)^{|I|} \int_D \partial_I (E_a R_\alpha^{aI}) \varepsilon^\alpha \,\mathrm{vol}.$$

At every step the discarded total derivative contains a derivative of ε as a factor and has compact support in U. Extend it by zero and integrate over $\mathbb{R}^{1+d}$. Fubini and the ordinary fundamental theorem make its integral zero. No boundary regularity theorem for D is needed. Summing the finite list of terms proves the pairing identity

$$0 = \delta S_D[\phi] R_\phi(\varepsilon) = \int_D E_a R_\phi(\varepsilon)^a \,\mathrm{vol} = \int_D (R_\phi^* E)_\alpha \varepsilon^\alpha \,\mathrm{vol}.$$

The parameter components are arbitrary smooth compact tests. The smooth-test fundamental lemma of Lecture 9 therefore gives $(R_\phi^* E)_\alpha = 0$ on U for every component. The field was arbitrary, so this is an off-shell identity. ∎

The theorem concerns infinitesimal invariance under arbitrary functions. It does not require the field-dependent coefficients to be constant, nor a finite-dimensional parameter space as in Lecture 2.

19.3 The pure Yang–Mills identity

We now work on $\mathbb{R}^{1+3}$. We recall the needed matrix conventions from *Topology and Geometry*, Part II, Lectures 21 and 23, and verify the identities used here. Let $A = A_\mu \mathrm{d}x^\mu$ be a smooth one-form with values in $\mathfrak{su}(n_c)$, the anti-Hermitian traceless $n_c \times n_c$ matrices, where $n_c \ge 2$. The curvature, covariant derivative and real invariant pairing are

$$\begin{aligned} F_{\mu\nu} &= \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu], \\ D_\mu X &= \partial_\mu X + [A_\mu, X], \qquad \langle X, Y\rangle = -2\,\mathrm{tr}(XY). \end{aligned} \tag{19.4}$$

Here $[X,Y]=XY-YX$. In form notation $F=\mathrm{d}A+A\wedge A$; matrix multiplication accompanies the wedge product. On forms, $D\omega=\mathrm{d}x^{\mu}\wedge(\partial_{\mu}\omega+[A_{\mu},\omega])$, with the commutator taken in each coefficient. The metric and orientation are the Minkowski ones of Lecture 9.

The pairing is nondegenerate because $\langle X,X\rangle=2\operatorname{tr}(X^{\dagger}X)>0$ for $X\neq 0$. In a real orthonormal basis for this pairing, it is the sum of products of real components. Thus the component form of Theorem 19.3 applies to these matrix-valued fields and parameters. Trace cyclicity gives $\langle[A,X],Y\rangle+\langle X,[A,Y]\rangle=0$. Thus the product rule takes the covariant form

$$\partial_{\mu}\langle X,Y\rangle=\langle D_{\mu}X,Y\rangle+\langle X,D_{\mu}Y\rangle.$$

For compactly supported parameters this makes the formal adjoint of $R(\varepsilon)_{\mu}=D_{\mu}\varepsilon$ equal to $R^{*}E=-D_{\mu}E^{\mu}$.

Fix a coupling $g>0$. The *pure Yang–Mills action* is

$$S_D[A]=-\frac{1}{4g^2}\int_D\langle F_{\mu\nu},F^{\mu\nu}\rangle\,\mathrm{vol}. \tag{19.5}$$

Spacetime indices are raised with η. Under a compact variation a_{μ}, direct differentiation gives $\delta F_{\mu\nu}=D_{\mu}a_{\nu}-D_{\nu}a_{\mu}$. Antisymmetry and the covariant product rule then give

$$\begin{aligned}\delta S_D[A]a&=-\frac{1}{g^2}\int_D\langle D_{\mu}a_{\nu},F^{\mu\nu}\rangle\,\mathrm{vol}\\&=\frac{1}{g^2}\int_D\langle a_{\nu},D_{\mu}F^{\mu\nu}\rangle\,\mathrm{vol},\qquad E^{\nu}=\frac{1}{g^2}D_{\mu}F^{\mu\nu}.\end{aligned} \tag{19.6}$$

The discarded divergence is $-g^{-2}\partial_{\mu}\langle a_{\nu},F^{\mu\nu}\rangle$, which has compact support. The equations $D_{\mu}F^{\mu\nu}=0$ are written $D{\star}F=0$ in form notation.

The gauge variation $a_{\mu}=D_{\mu}\varepsilon$ gives $\delta F_{\mu\nu}=[F_{\mu\nu},\varepsilon]$. To verify this, expand two covariant derivatives and use the matrix Jacobi identity:

$$\begin{aligned}[D_{\mu},D_{\nu}]X&=[\partial_{\mu}A_{\nu}-\partial_{\nu}A_{\mu},X]+[A_{\mu},[A_{\nu},X]]-[A_{\nu},[A_{\mu},X]]\\&=[F_{\mu\nu},X].\end{aligned}$$

The variation of $\operatorname{tr}(F_{\mu\nu}F^{\mu\nu})$ is the trace of $[F_{\mu\nu}F^{\mu\nu},\varepsilon]$, hence zero. Thus the action has this gauge symmetry, and Theorem 19.3 gives the pure Yang–Mills identity. It can also be checked directly:

$$2D_{\mu}D_{\nu}F^{\nu\mu}=[D_{\mu},D_{\nu}]F^{\nu\mu}=[F_{\mu\nu},F^{\nu\mu}]=0. \tag{19.7}$$

The first equality interchanges the dummy indices and uses antisymmetry. For each index pair, $F^{\nu\mu}$ is a scalar multiple of $F_{\mu\nu}$ in our diagonal metric, so the final commutator is zero. This is an identity for every smooth connection, without its equations. For coupled matter, the gauge operator also acts on the matter fields; their Euler expressions then enter the Noether identity. Sheet 19 treats that extension.

19.4 Maxwell charge on a sphere

Return to the Maxwell normalisation of Lecture 9. At fixed time, its Hodge table pulls $\star F_{\mathrm{em}}$ back to $E_x\mathrm{d}y\wedge\mathrm{d}z+E_y\mathrm{d}z\wedge\mathrm{d}x+E_z\mathrm{d}x\wedge\mathrm{d}y$. Give $S^2_r=\partial B_r(0)$ the outward boundary orientation of spatial $\mathbb{R}^3$. The ball divergence formula proved by coordinate FTC in Lecture 5 gives

$$\int_{S^2_r}\star F_{\mathrm{em}}=\int_{S^2_r}\mathbf{E}\cdot n\,\mathrm{d}S=\int_{B_r}\operatorname{div}\mathbf{E}\,\mathrm{d}^3x=\int_{B_r}\rho\,\mathrm{d}^3x. \tag{19.8}$$

The form is pulled back to the sphere in a constant-time slice. For compactly supported charge and a smooth Maxwell field, the right side is the total charge Q once the sphere contains the support. Consequently $Q = \lim_{r\to\infty} \int_{S^2_r} \star F_{\mathrm{em}}$. Figure 19.1 records the orientation.

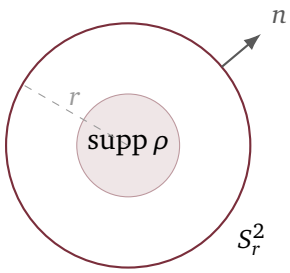


Figure 19.1: A spatial section of a sphere enclosing the charge support. The arrow gives the outward normal used for the electric flux.

Example 19.4 (A smooth static charge with a Coulomb tail)**.** Choose a smooth radial density $\rho(x) = \rho_0(|x|)$ supported in $B_L(0)$, with total charge $Q \neq 0$. Take $\mathbf{J} = 0$ and $\mathbf{B} = 0$, and define

$$M(r) = 4\pi \int_0^r s^2 \rho_0(s)\, \mathrm{d}s, \qquad \mathbf{E}(x) = \frac{M(r)}{4\pi r^3} x \quad (r = |x| > 0).$$

The identity $\mathbf{E}(x) = x \int_0^1 a^2 \rho(ax)\, \mathrm{d}a$ extends the field smoothly through $x = 0$. Differentiating that expression gives $\operatorname{div} \mathbf{E} = \int_0^1 \partial_a [a^3 \rho(ax)] \mathrm{d}a = \rho(x)$. The curl vanishes because the field is radial. These time-independent fields therefore solve all Maxwell equations. Their scalar potential can be chosen as

$$\phi_{\mathrm{em}}(r) = \int_r^\infty \frac{M(s)}{4\pi s^2}\, \mathrm{d}s, \qquad \mathbf{A} = 0.$$

It has $-\nabla \phi_{\mathrm{em}} = \mathbf{E}$ and is smooth at the origin since the electric field is smooth there. For $r > L$, it is exactly $Q/(4\pi r)$. The outward flux is $4\pi r^2 E_r = M(r)$, equal to Q outside the source. The potential has a non-L^2 Coulomb tail. This static charge persists for all times, so it is outside Lecture 10's compact-spacetime source class. It is a regular extended charge, with no point singularity at the origin.

19.5 Covariant divergence cannot be replaced by divergence

The Yang–Mills identity says $\partial_\mu E^\mu = -[A_\mu, E^\mu]$, which need not vanish. For example, in a spatial coordinate patch take $\mathfrak{su}(2)$ generators $T_a = -\mathrm{i}\sigma_a/2$, with $[T_1, T_2] = T_3$ cyclically, and static potentials $A_1 = T_1$, $A_2 = xT_2$, $A_0 = A_3 = 0$. For $g = 1$, direct differentiation gives

$$E^1 = -x^2 T_1, \qquad E^2 = -xT_2 + 2T_3, \qquad \partial_\mu E^\mu = -2xT_1.$$

The commutator $[A_2, E^2] = 2xT_1$ restores the zero covariant divergence. This is a local off-shell example, not a finite-energy assertion on all space.

The temporal Euler equation gives the non-abelian Gauss law. With $\Pi_i = g^{-2} F_{0i}$, the diagonal metric gives

$$E^0 = D_i \Pi_i = \partial_i \Pi_i + [A_i, \Pi_i] = 0. \tag{19.9}$$

The commutator must be retained. Lecture 20 studies its Hamiltonian evolution.

Summary. For a finite-order parameter-dependent symmetry, repeated coordinate integration by parts gives $R^*E = 0$ off shell. Derivatives act on the operator's evaluated coefficients as well as on E. For pure Yang–Mills theory the formal adjoint is the negative covariant divergence, giving $D_\mu D_\nu F^{\nu\mu} = 0$. Maxwell charge is the outward flux of $\star F_{\mathrm{em}}$ through a large sphere. The non-abelian Gauss equation contains a commutator, so ordinary divergence alone does not express the constraint.

Self-check. In one coordinate, let $R(\varepsilon) = a(x)\varepsilon_{xx} + b(x)\varepsilon_x$. What is R^*E for compactly supported parameters?

Answer. The result is $\partial_x^2(aE) - \partial_x(bE)$. Both derivatives act on the coefficients. Dropping them would change the integration-by-parts identity when a or b varies.

Self-check. Does $R^*E = 0$ require solving $E = 0$ first? What changes if gauge variations also act on matter fields?

Answer. It holds at every smooth field, before imposing any equation. For matter fields, the first variation and gauge operator have additional components, so the identity includes the corresponding matter Euler expressions.

Self-check. For the static radial Maxwell example, reverse the orientation of the large sphere. What flux is obtained? Why is there no factor 4π multiplying Q?

Answer. The reversed flux is $-Q$. Our field outside the source is $Qn/(4\pi r^2)$, so the area $4\pi r^2$ cancels the denominator. This uses the rationalised Maxwell normalisation.

Self-check. Choose a smooth compact cutoff $\vartheta = 1$ near the origin and put $f = y\vartheta$. Take $A_1 = \vartheta T_1$, $A_2 = A_3 = 0$ and $\Pi_1 = f_y T_2$, $\Pi_2 = -f_x T_2$, $\Pi_3 = 0$. Compare the ordinary and covariant Gauss expressions at the origin.

Answer. Mixed derivatives give $\partial_i \Pi_i = 0$ everywhere. The commutator gives $D_i \Pi_i = \vartheta f_y T_3$, equal to T_3 at the origin. All arrays are smooth and compactly supported. The ordinary equation alone would accept these data even though the non-abelian Gauss constraint fails.